

TINKER: A Fully-Digital Threshold-Logic-Mapped BNN Inference SoC

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85.60 %

MNIST accuracy
full 10 000-image test set
95 % Wilson CI [84.90, 86.27]

8.06 μ J

per inference
17 068 cycles, 170.7 μ s
at 100 MHz

47.21 mW

post-P&R SoC power
activity-aware, 2.25 mm² die,
sky130A open flow

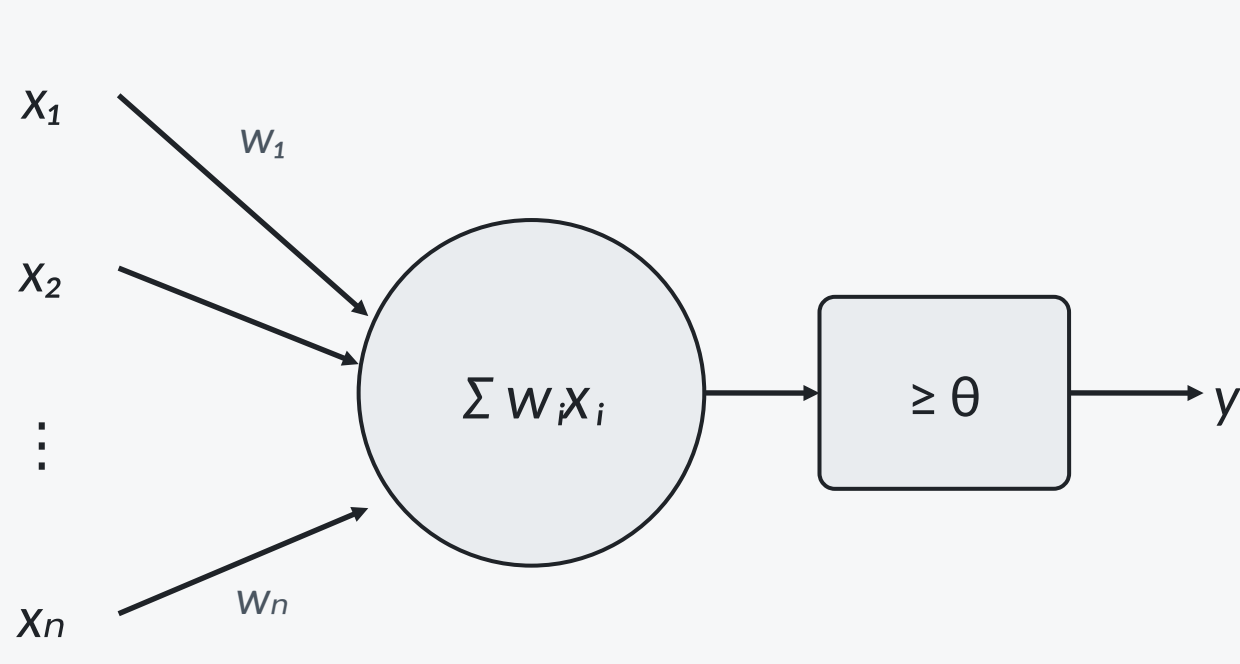
24.3 %

of SoC power in the TLG tile
largest active datapath bucket

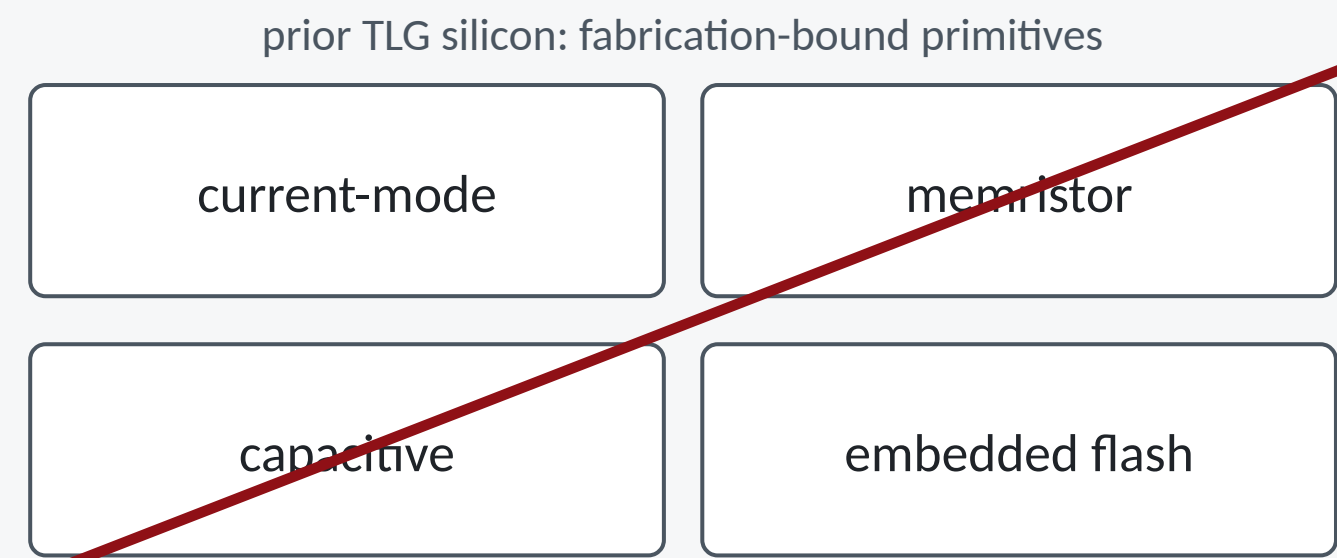


paper + poster (Drive)

1 Threshold neuron to standard cells



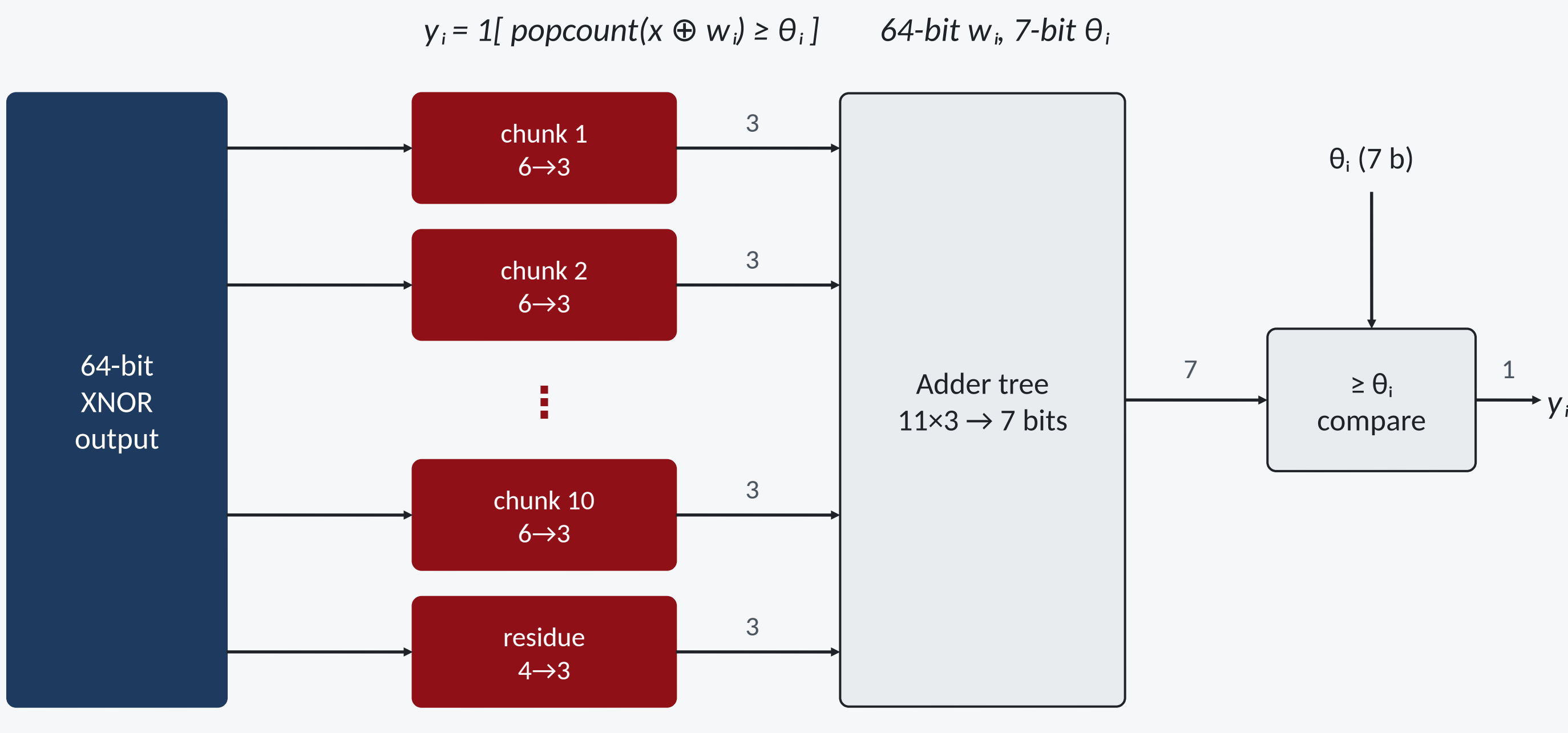
Threshold-logic neuron: fires when a weighted sum crosses θ



sky130_fd_sc_hd standard cells only

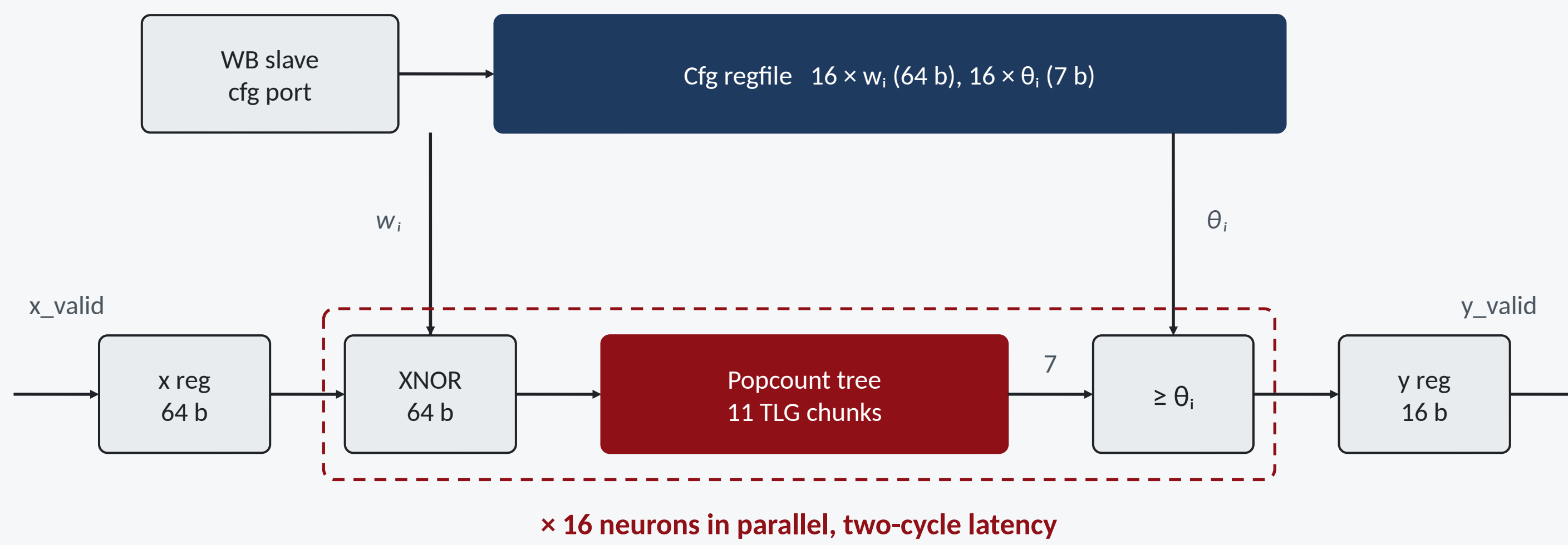
Can a threshold-logic BNN live in an ordinary digital RTL-to-GDS flow, inside a complete SoC?

2 Chunked-LUT threshold decomposition



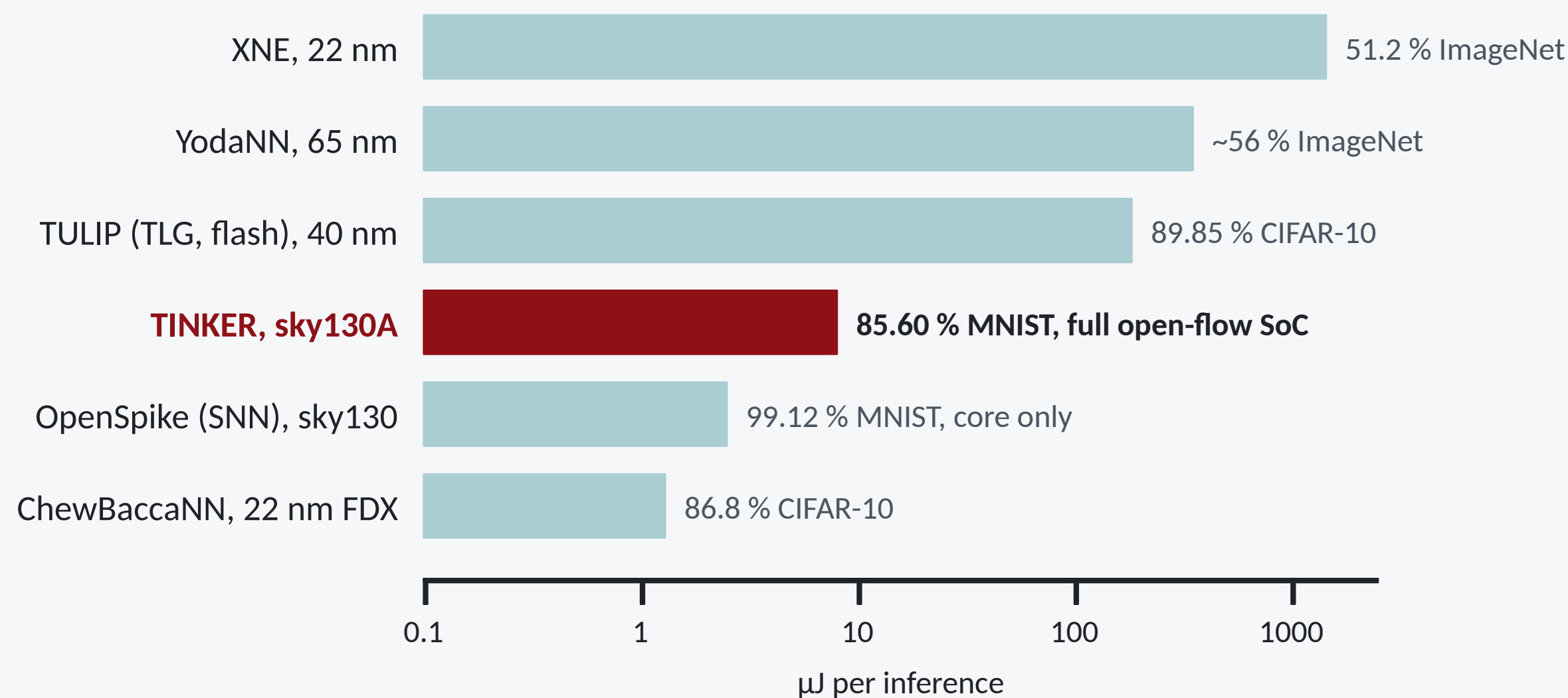
sky130_fd_sc_hd standard cells only (muxes, AOI cells, adders)

3 Sixteen-neuron tile, two-cycle latency



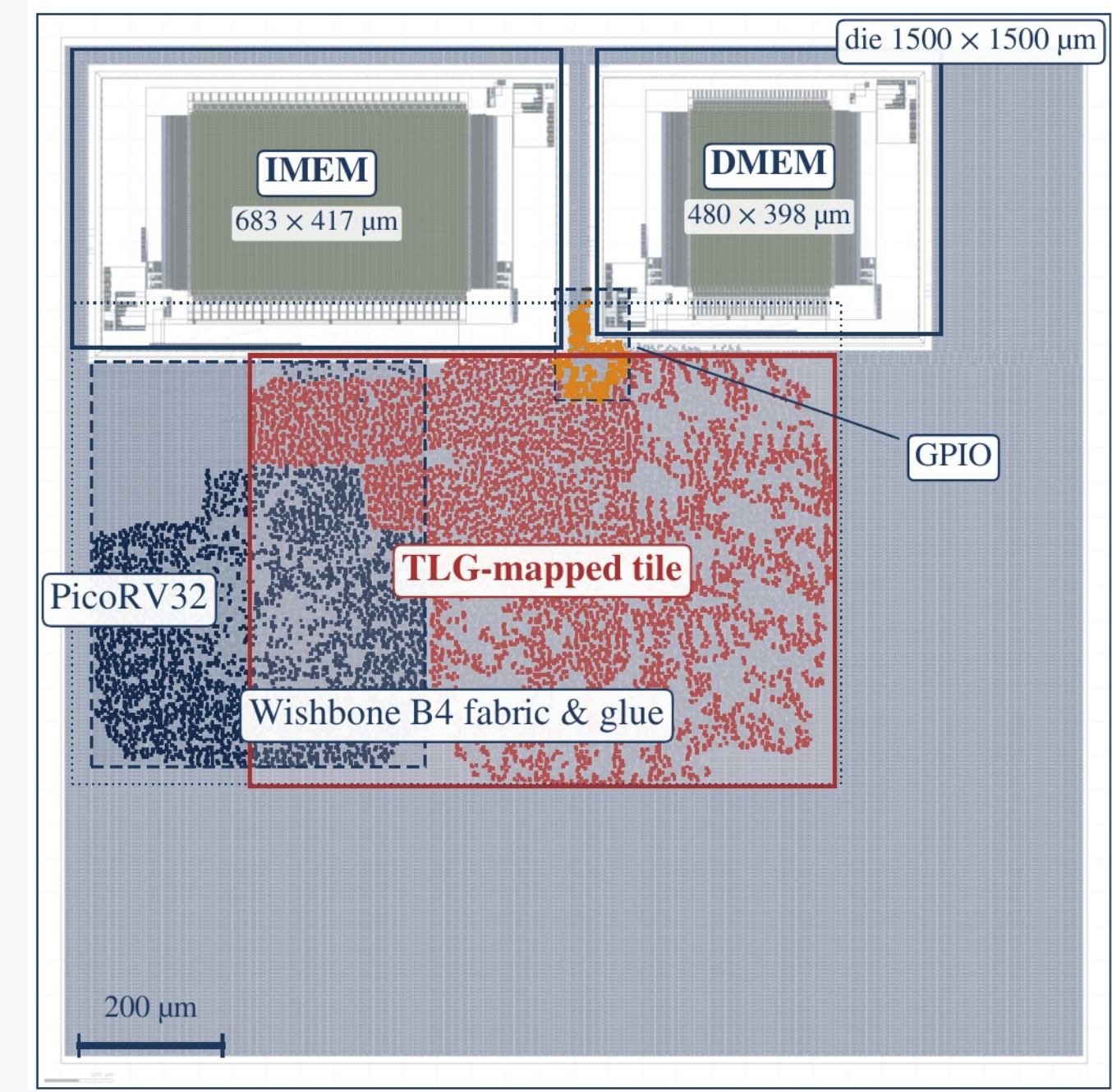
× 16 neurons in parallel, two-cycle latency

4 Energy per inference vs. prior BNN silicon (log scale)



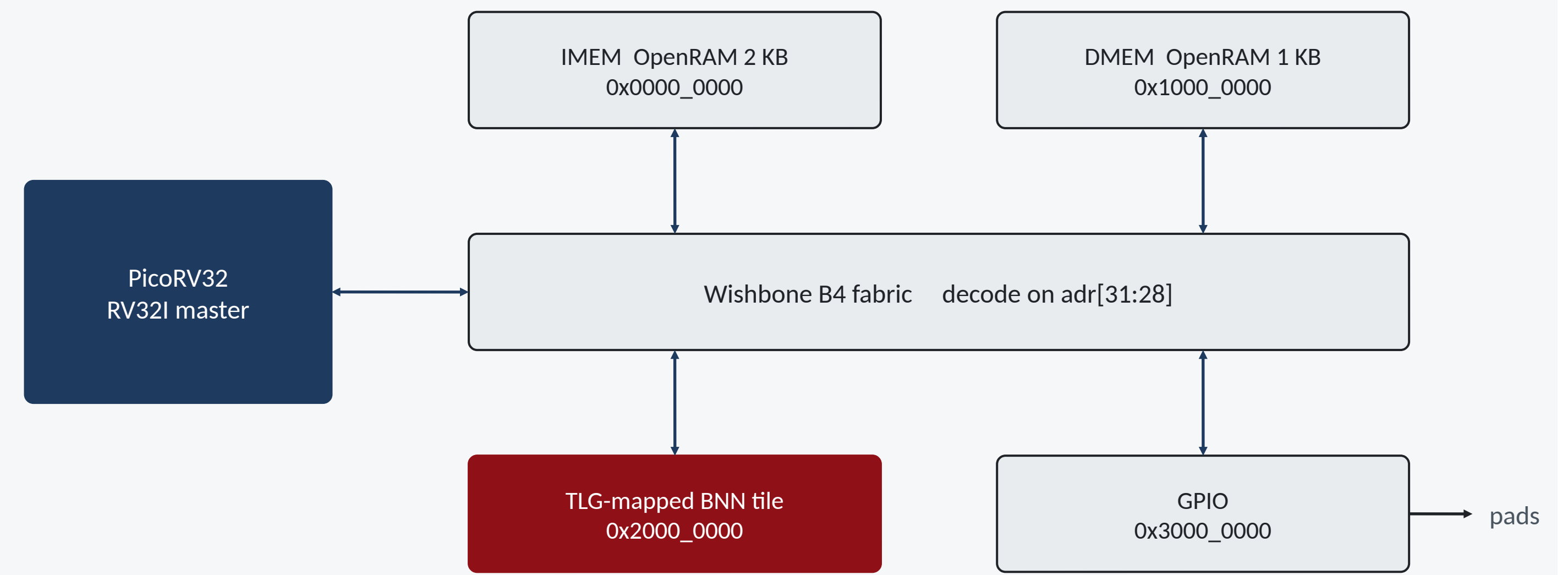
Nodes, networks and reporting scope differ; positioning, not a ranking. 124 113 inferences/J, 5 860 inf/s.

5 Post-P&R floorplan on sky130A, 1.5 × 1.5 mm die

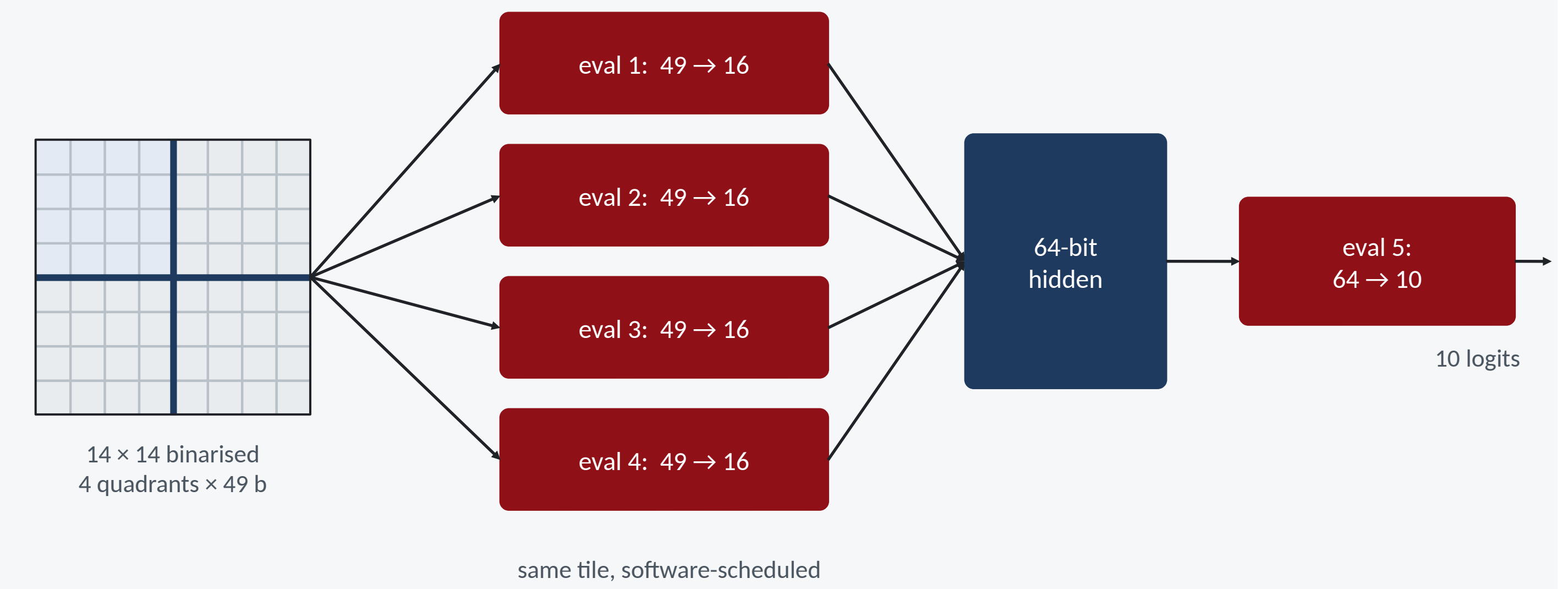


LibreLane flow: Yosys, OpenROAD, Magic, KLayout, Netgen. OpenRAM IMEM (2 KB) and DMEM (1 KB) macros top; TLG tile, PicoRV32, and fabric below. Nine-corner STA; TT closes at +0.51 ns setup, +0.37 ns hold.

6 One PicoRV32 master, four Wishbone B4 slaves, 100 MHz

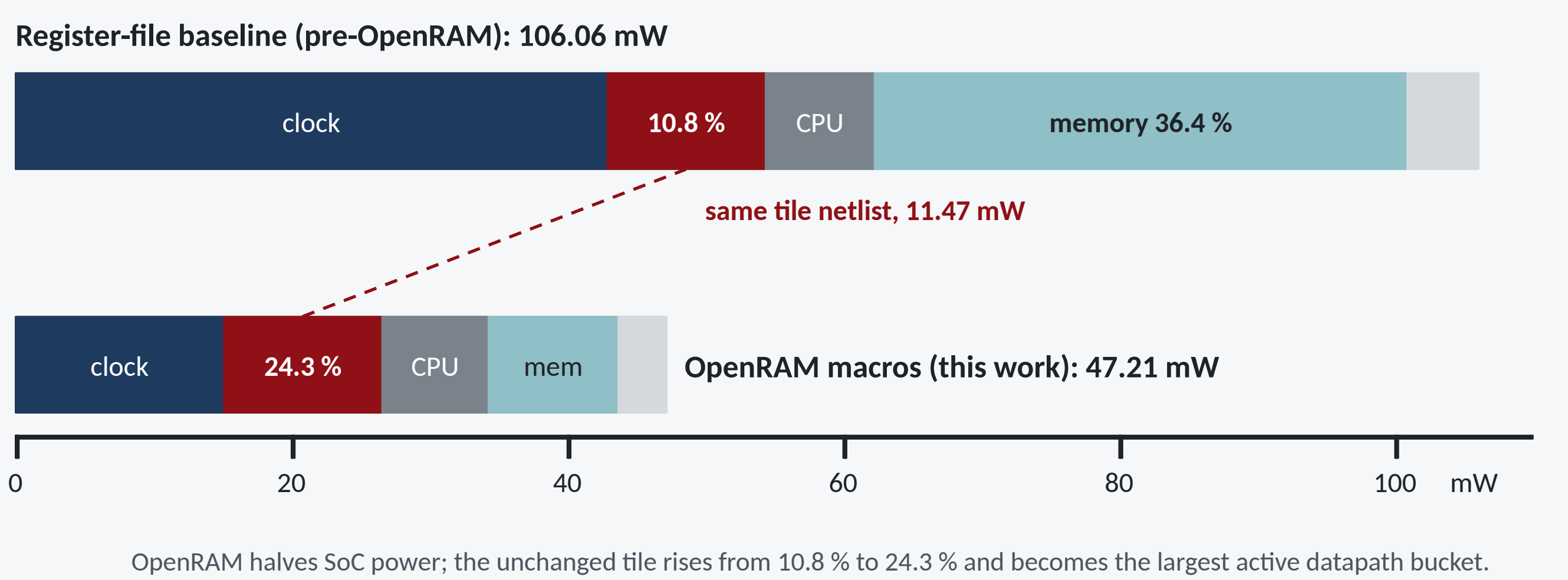


7 Grouped 14 × 14 MNIST BNN: five tile evaluations per image



same tile, software-scheduled

8 Where the SoC power goes (post-P&R, VCD-annotated)



OpenRAM halves SoC power; the unchanged tile rises from 10.8 % to 24.3 % and becomes the largest active datapath bucket.